Linear Algebra

Joe Razavi

20th March 2025

Chapter 9

Linear Algebra

9.1.1 Vector spaces over the real numbers

Definition 68: Vector space over the real numbers

A **vector space over the real numbers** (often shoertened to "real vector space") is a set V together with

- A binary operation $+ : V \times V \to V$ called "vector addition",
- An element $\underline{0} \in V$
- A function from $\mathbb{R} \times V$ to V called "scalar multiplication". For $r \in \mathbb{R}$ and $\underline{v} \in V$ we simply write $r\underline{v}$ to mean the scalar multiplication of r and $\underline{v}$

satisfying the following properties:

- $+$ is commutative and associative
- $\underline{0}$ is a unit for $+$
- every element $\underline{v}$ of V has an inverse for the operation $+$ which we write as $-\underline{v}$

And for all $r, s \in \mathbb{R}$ and $\underline{v}, \underline{w} \in V$ we have

- $1\underline{v} = \underline{v}$, we say that 1 is a **scalar unit**
- $r(s\underline{v}) = (rs)\underline{v}$, which we call **scalar compatibility**
- $(r + s)\underline{v} = r\underline{v} + s\underline{v}$, which we call **scalar distributivity**
- $r(\underline{v} + \underline{w}) = r\underline{v} + r\underline{w}$, which we call **vector distributivity**

Note that by convention we write a vector with an underscore – different authors use different conventions so take care. The names of the various properties are not completely standard, so you may find variations in difference sources. We shall also use the common convention of writing $\underline{v} - \underline{w}$ to mean $\underline{v} + (-\underline{w})$.

These properties give rise to a number of other useful properties.

Theorem 9.1

Let V be a vector space over the real numbers. Then for all $\underline{v}, \underline{w} \in V$ and $r, s \in \mathbb{R}$ we have

- $0\underline{v} = \underline{0}$
- $r\underline{0} = \underline{0}$
- $(-1)\underline{v} = -\underline{v}$

Proof.

- We have

$$\begin{aligned}
0\underline{v} &= 0\underline{v} + \underline{0} && \\
&= 0\underline{v} + (\underline{v} - \underline{v}) && \text{def. unit} \\
&= (0\underline{v} + \underline{v}) - \underline{v} && \text{def. inverse} \\
&= (0\underline{v} + 1\underline{v}) - \underline{v} && \text{assoc. } +\text{, scalar unit} \\
&= (0 + 1)\underline{v} - \underline{v} && \text{scalar distributivity} \\
&= 1\underline{v} - \underline{v} && \text{calculation in } \mathbb{R} \\
&= \underline{v} - \underline{v} && \text{scalar unit} \\
&= \underline{0} && \text{def. inverse}
\end{aligned}$$

- We have

$$\begin{aligned}
r\underline{0} &= r(\underline{0} + \underline{0}) && \text{def. unit} \\
&= r\underline{0} + r\underline{0} && \text{vector distributivity}
\end{aligned}$$

  So adding the inverse of $r\underline{0}$ to both sides we have

$$r\underline{0} - r\underline{0} = r\underline{0} + r\underline{0} - r\underline{0}$$

  and hence by the definition of inverse

$$\underline{0} = r\underline{0} \text{ as required}$$

- See exercise below!

CExercise 196. (on Sheet 18) Prove the final part of the theorem above, that is show that for all $\underline{v} \in V$ we have $(-1)\underline{v} = -\underline{v}$. You can make use of everything proved or stated in the notes up to and including the definition of a vector space, as well as the first two parts of the theorem.

Exercise 197. Can you find a shorter proof of the first statement in the theorem to the one given? *Hint: start by considering the fact that* $0\underline{v} = (0 + 0)\underline{v}$

Let's look at some boring examples first, to get an idea what this all means.

Example 9.1. The set $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ is a vector space over the reals when equipped with the following operations:

$$(x, y) + (x', y') = (x + x', y + y')$$

$$r(x, y) = (rx, ry)$$

$$\underline{0} = (0, 0)$$

To show that this is a vector space, we have to check all of the properties from the definition.

Example 9.2. The real numbers $\mathbb{R}$ themselves form a vector space over the reals. For the operation $+$ we use ordinary addition, and for scalar multiplication we can use ordinary multiplication. This example is a bit confusing, because we usually take care to distinguish between addition and multiplication in $\mathbb{R}$ and vector addition and scalar multiplication in a real vector space. However, in this example (only!) they coincide.

Exercise 198. Show that $\mathbb{C}^2$ is a vector space over the reals, when equipped with the following:

- $\underline{0} = (0, 0) \in \mathbb{C}^2$
- $(x_0, x_1) + (z_0, z_1) = (x_0 + z_0, x_1 + z_1)$
- $r(x_0, x_1) = (rx_0, rx_1)$

Exercise 199. Determine whether $\mathbb{C}^2$ is a vector space over the reals, when equipped with the following:

- $\underline{0} = (0, 0) \in \mathbb{C}^2$
- $(x_0, x_1) + (z_0, z_1) = (x_0 + z_1, x_1 + z_0)$
- $r(x_0, x_1) = (rx_0, rx_1)$

Prove that your answer is correct.

Exercise 200. Let X be a set. Determine whether the set of functions from X to $\mathbb{R}$ form a vector space over the reals when, for all $x \in X$ we define the operations of the vector space to have the following behaviour.

- $\underline{0}(x) = 0$
- $(\underline{f} + \underline{g})(x) = \underline{f}(x) + \underline{g}(x)$
- $(r\underline{f})(x) = r(\underline{f}(x))$

Prove that your answer is correct

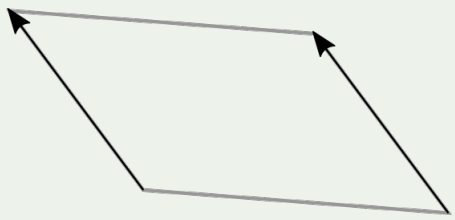
Conceptual examples of vector spaces

As with many of the topics we study on this unit, we're also interested in thinking conceptually about examples of vector spaces we don't define: those which come from real world situations, or from area of geometry or other kinds of maths that we don't study in detail. Thinking about this is important to show how widely the field of linear algebra can be applied, but naturally you won't be able to get a complete rigorous understanding of them—for many I don't have a clear understanding myself! Still, it is a good idea to ponder them to help expand your concept of what a "vector" can be.

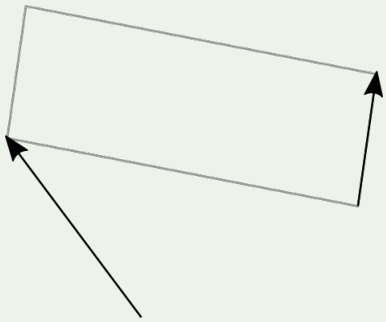
Many of these in fact correspond in some way to the "ordinary" notion of vector you may have seen in school—but we discuss them with greater conceptual care to bring out subtle aspects which are usually swept under the rug. Spend as much time as you find interesting pondering these—but don't really try to get to understand them. Once you feel that the concept of vector space is more subtle than you might have thought, you can move on, and perhaps return to them in the summer, once you have all the mathematical tools of the course unit at your disposal!

Example 9.3. In 2D Euclidean geometry, we can define a notion of "vector" which is a relationship which can hold between an ordered pair of points. Mathematically, we define what we mean by giving a set of geometric objects and an equivalence relation between them, and the set we want is the quotient set.

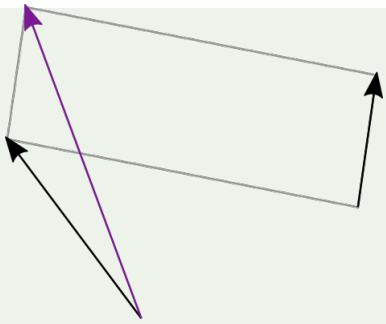
The geometric objects are just ordered pairs (p_0, p_1) of points of Euclidean space. We usually draw them as little arrows: line segments marked with an arrow tip pointing at p_1. We consider two of these to be equivalent if they form two opposite sides of a parallelogram:



Now given a pair of equivalence classes $\underline{v}$ and $\underline{w}$, and representative elements (v_0, v_1) of $\underline{v}$ and (w_0, w_1) of $\underline{w}$, we can construct a parallelogram such that one of its sides goes from v_0, the first point of the element of $\underline{v}$, to w_1, the end point of the element of $\underline{w}$:



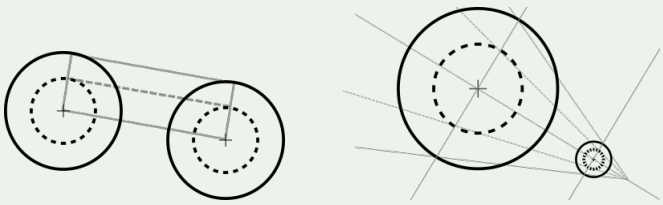
Now we define the sum to be the equivalence class of the highlighted arrow below:



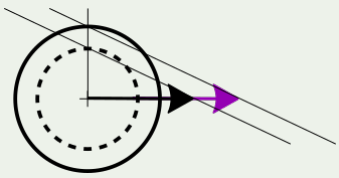
Namely the pair of points whose first point is w_0 and whose endpoint is the corner of the parallelogram opposite v_1. We have to check that this operation always outputs equivalent elements when given equivalent elements as inputs (otherwise it doesn't even define a binary operation on the quotient set!). Then we have to check it is associative and commutative, and has a unit element (given by two points in the same place). You can imagine this is quite laborious if we do it all strictly using Euclidean geometry.

But scaling by a real number is even more sophisticated! So much so that I won't go into great detail here. We can first define scalar multiplication of -1 with any Euclidean vector to be reverse of the original. Now we just have to define the scalar multiplication with any positive real number. Of course intuitively this is a line segment pointing in the same direction but which is longer by the given scale factor.

However, if we wanted to start from first principles, we would need to define "longer by a given scale factor" geometrically. One possible idea is to represent a real number by an equivalence class of pairs of concentric circles (intuitively a real r is represented by the set of pairs of circles where the radius of the second is r times greater than that of the first). To define this equivalence geometrically we consider circle-pairs equivalent *both* when they can be translated using parallelograms, like vectors, and when they are related by a scaling. Here these are depicted sketchily by showing the first circle in each pair with a dotted line.



You might like to think how addition can be defined for this representation of real numbers; we now turn to the definition of scalar multiplication of a vector which should be a clue for how multiplication can be defined. Given a representative of a vector $\underline{v}$ and a real number r, we use a translation to move the centre of the circles to the first point of the vector, then construct a line perpendicular to the representative of $\underline{v}$, and then two parallel lines like this:



One of the parallel lines passes through the endpoint of the representative of $\underline{v}$ and the intersection of the perpendicular and the dotted circle. The other is parallel to it and passes through the intersection of the perpendicular and the solid circle. The common centre of the circles and the point where the second parallel line crosses the line of the representative of $\underline{v}$ form the representative of the scaled version. It is now an exercise in Euclidean geometry to show that this does not depend on the representatives of the vector *or* the real number chosen, so that it successfully defines an operation on the quotient sets.

Given how involved all this is just to *define*, you can imagine the remaining effort needed to show that this is indeed a vector space from the axioms of Euclidean geometry!

The example above is interesting for several reasons. First, you may feel that this *is* the vector space $\mathbb{R}^2$, and indeed nothing we do in linear algebra can tell them apart. However, you would get stuck if I asked you which element of $\mathbb{R}^2$ (that is, which pair of real numbers) corresponds to any of the vectors drawn in the example above! To assign a pair of numbers to one of these vectors, we would have to choose a co-ordinate system and a unit of length, and there is no preferred choice. As we vary the choice of co-ordinate system, the particular pair of numbers varies, leading to the physicist's view of a vector as a function from co-ordinate systems to lists of numbers which varies in a particular way. This is a bit like regarding your height as a procedure for taking a unit of length and returning a real number!

The second interesting thing is that Euclidean space has much more structure than the vector space we construct. For one thing, in Euclidean space the angle between two line segments can be measured, and the lengths of lines in different directions can meaningfully be compared. We actually use these properties to construct circles and perpendicular lines in the definition. However, if we forget everything about the geometric origin of this vector space and work only with the operations of vector addition and scalar multiplication, we can *never recover* the notion of the angle between vectors, or the idea that vectors which point in different directions could have the same length. The vector space structure simply doesn't contain this information! In this sense you might think of a vector space as a slightly "wobbly" structure, compared to the rigid world of Euclidean geometry.

Example 9.4. In the above, translations and scalings play an important role. Indeed, we can define a vector space whose vectors are translations of Euclidean space, and where vector addition is just doing one translation after another. To define the scalar product $r\underline{v}$ we make use of the scaling transformation s_r which scales everything up by a factor of r, and its inverse $s_{r^{-1}}$. Then for a translation τ we define

$$r\underline{\tau} = s_r \circ \tau \circ s_{r^{-1}}$$

To get an intuition for why this works, you might like to think of the simpler case where we represent a real number t by the function $\tau_t : \mathbb{R} \to \mathbb{R}$ which "translates" the real line by adding t to its argument. Then a scaling s_r is just the function which multiplies its argument by r, and for all $x \in \mathbb{R}$ we have

$$(s_r \circ \tau_t \circ s_{r^{-1}})(x) = s(x/s + t) = x + st$$

Unlike $\mathbb{R}^2$, this vector space of translations is very closely related to the Euclidean vector space, because for any ordered pair of points there is a *unique* choice of a translation which maps one to the other. However, the mathematical objects involved (translation functions vs pairs of points) are still clearly different in the sense that we might prefer doing one way to the other. Proving the translation version is a vector space would involve doing lots of geometry to establish various properties of translation and scaling operations, which would allow us to

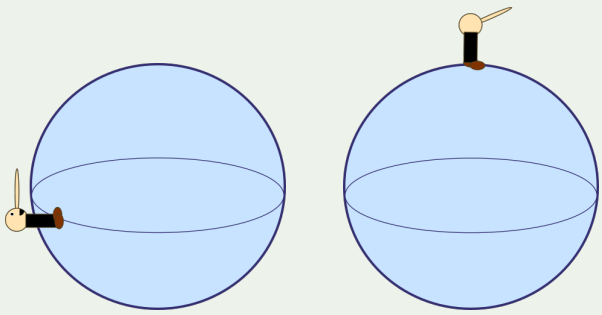
reason about the more elegant-looking definitions of vector addition and scalar multiplication in terms of composition of functions and inverses.

Example 9.5. The set of velocities through a point in Newtonian mechanics forms a vector space. The addition of velocities can be found by working out the velocity of an object travelling at velocity $\underline{v}$ relative to an object travelling at velocity $\underline{w}$, like someone running on the deck of a boat. The velocity of the runner from the point of view of a stationary observer on the shore would be the vector addition of their velocity accross the deck with that of the boat accross the water. The scalar multiple of a velocity $\underline{v}$ by r is the velocity with the same direction as $\underline{v}$ but with a speed increased by a factor of r (note the direction will be opposite if r is negative) – or the velocity that an object travelling with velocity $\underline{v}$ would seem to travel at in a movie which as been sped up by a factor of r.

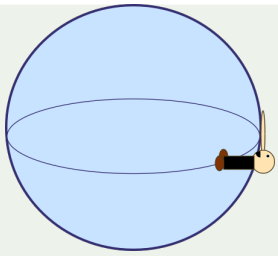
We might feel, again, that this is "the same" as the Euclidean vector space. If we ignore the third dimension and think about 2D velocities, then we may feel that like Euclidean vectors, they can be described by pairs of real numbers. However, to do this requires that we choose not only a set of co-ordinates and units for length, but *also units of time*. What we are seeing here is that for some kinds of reasoning, it is convenient to pick co-ordinates and units and work with lists of numbers, but for others it might be better to stay aloof from calculation and treat different kinds of quantities as really living in fundamentally different vector spaces. Converting everything to numbers, and thus making distances, times, and many other things appear to be the same kind of data, is one place where bugs can creep into our code!

Example 9.6. (Non-example!) Can we repeat what we did for Euclidean space for the surface of a sphere? We know that in real life if we walk "straight ahead without turning", or "following our nose", over a long distance, we actually follow a circular path because gravity keeps us fixed on the surface of the Earth. These "geodesic" paths where we don't feel we are turning have a very different geometry to the straight lines of Euclidean space, and the geometric construction above goes wrong! Another way to see this is that instead of translations, the operations we have on a sphere correspond to 3D rotations, which are much more complicated—in particular their composition is not commutative! We shall return to this later once we have the mathematical machinery to discuss it, but for now we can sketch the kind of problem we face.

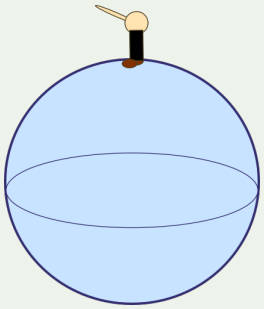
Consider a person with a very long nose standing on the equator, and facing north. Consider what happens if they walk in what feels like a "straight line", until they eventually reach the North Pole:



But if someone carries them east a good distance first, again carrying them in what feels like a straight line, without seeming to turn them, we get to this position:



But now if they walk "straight" to the North Pole, they end up like this:



What this means is that when travelling large distances over the Earth, we have operations which feel like translations—moving things without rotating them–but which can go between the same two places with different results. This does not yet formally show that a vector space structure can't be defined from these "straight" paths over a sphere, but somewhat illustrates the problem because we used operations like this to define vector addition. Whatever we might try, we either get a non-commutative addition or, even worse, when we use different representatives of the same equivalence class, we get very different answers depending on their relative locations on the sphere.

Example 9.7. In contrast, if we assume a Newtonian account of physics, the velocities of objects through a given point on the Earth's surface *do* have a vector space structure! We can add them as we added velocities before: imagine the velocity of an object which seems to have velocity $\underline{v}$ when observed from an object with velocity $\underline{w}$. It's velocity relative to the ground will be $\underline{v} + \underline{w}$, and we can scale by speeding up the velocity as usual. The reason this works is that over very short distances, the surface of the Earth is very well approximated near any given point by a flat plane, and since velocities can be measured, at least conceptually, over arbitrarily short times, we can use the Euclidean vector space structure on the plane to make sense of what adding and scaling velocities mean.

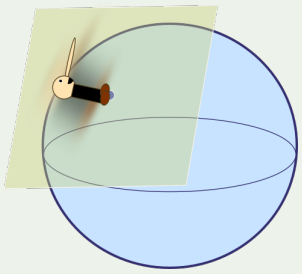


Figure: in a very short exposure photo, the motion blur of someone rushing past a point on the Earth approximately lies along the plane touching the surface at that point.

Note that this idea gives us a *different vector space for each point* on the Earth, because the cartoon in the previous example shows there is no sensible way of saying that a velocity vector at one point of the Earth is pointing in the same direction as a velocity somewhere else! Of course we can come up with conventions, such as our use of North, South, East, and West. But these conventions are badly behaved: at the North Pole everywhere is South, and when near the North Pole, you can feel that walking constantly East is not going in a straight line: you notice yourself turning—that is, "changing direction".

Example 9.8. For a final 2D example, suppose you are going on a hike, and to amuse yourself while resting, you have brought a flat booklet of squared paper, and a wind-up toy car. At home, you measured how many winds you need to make the car get from the middle of the page to each of the points. Now when you sit down you place the booklet on the hillside in front of you and measure again. You notice that the values are different from those you measured at home, and start looking into the extra winding you need to do to reach each point (negative extra winding for points which require less!).

After doing this in many locations, you observe that for every pair of points $\underline{v}$ and $\underline{w}$ there is a unique third point such that for *any* hillside, the extra winds needed to get to the third point is always the sum of the extra winds needed to get to $\underline{v}$ and the extra winds needed to get to $\underline{w}$. You call this point $\underline{v} + \underline{w}$. Similarly, for every real number r, there is a point which requires r times more extra winding to get to than $\underline{v}$, no matter where you put down the book, and you call this point $r\underline{v}$.

After playing this exciting game for many years, you notice that these definitions satisfy all the properties of a vector space. What is going on?

When you put down the booklet on a hillside, it acts like a plane approximating the undulating shape of the hill at that point. In different places, the booklet is sloping in different directions, and gravity either helps or hinders the toy car's motion in various directions. But because the booklet is a flat plane and we only conduct our experiments in a small area, the behaviour of this help or hindrance is simple and we can define the vector addition and scalar multiplication in terms of the results of our experiments.

Example 9.9. In modern physics velocity is a harder concept than in the Newtonian examples above because we can't add velocities by the same experiment of moving something inside a moving container. (Famously the speed of light is always fixed, so the speed you measure light coming out of an aeroplane headlights as it flies by is the same as out of your own torch sitting next to you).

However, physicists do make use of the fact that spacetime is approximately "flat" very close to any point, which means we can define a vector space structure on the approximate description. Since we no longer have an intuitive way to add velocities, the vectors in this vector space might be better thought of as deduced from experiments like the "toy car" example above. Various physical phenomena, like how waves move, depend on derivatives taken in spatial directions—directions no object could actually move in! From regularities observed in experiments which depend on these derivatives, we can define the vector addition and scalar multiplication operations with the correct behaviour, though these are now a bit abstract to think about. And similar to what we saw on the surface of a sphere, because spacetime is curved, each point has its own, unique vector space, and it doesn't really make sense to compare the vectors associated with distant points.

For a final conceptual example, let's turn back to geometry—this time to 3D space. In 3D Euclidean space, we can follow a similar geometric procedure as above to define geometric

vectors from pairs of points, using translations to say that two pairs of points represent the same vector if they are the same distance apart and point in the same direction. (Though our observations make us reflect that this notion of "distance" is *not* just a number, and the notion of direction is invisible to the resulting vector space!).

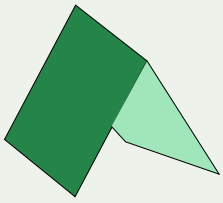
However, there are other ways of extracting a vector space from Euclidean geometry.

Example 9.10. In 3D Euclidean space, we can consider polygons: little flat surfaces. We want to consider "oriented" surfaces which you might imagine as very thin, flat objects which are different on either side (perhaps shiny on one side and dull on the other like a piece of kitchen foil). Mathematically there are various ways to represent this, for example by choosing an ordering of the vertices, which can be reversed to represent the "upside down" orientation of the polygon.

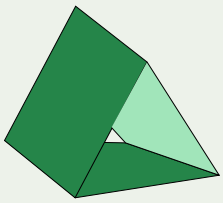
We put an equivalence relation on these by stating that they are the same if one can be split up into smaller polygons and rearranged into the other by translating these small pieces. It turns out that this means we care about the direction the surface is facing and the area of the surface, but not the position *or shape*.

We can scale such a thing by scaling up the area—we won't go into the Euclidean constructions which can do this for us. A negative scaling also flips the polygon over (exchanging the "shiny" and "dull" sides in our imagination)

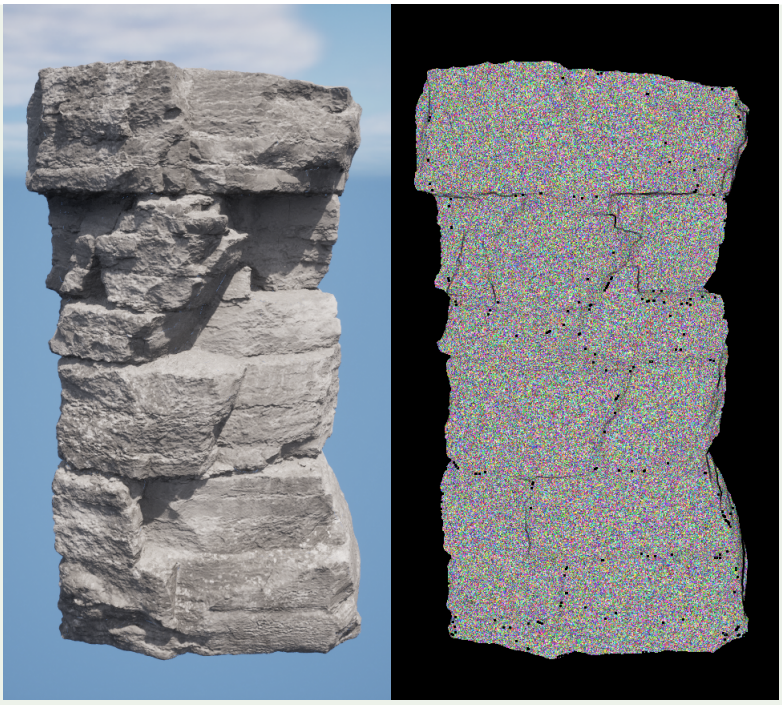
The addition of these polygonal vectors is a bit like that of ordinary "little arrow" vectors. First, since we don't care about their shape, we transform them into rectangles without changing the direction they face or their area. We then bring these rectangles together to form a "tent" shape with matching orientations (either the outside or the inside of the tent is all shiny, and the other side is all dull):



Now we fill in the "floor" of the "tent" with a matching orientation to find the sum of the two polygonal vectors:



It is hard to visualise, but these operations do form a vector space. This vector space is important in engineering applications, but also in computer graphics. For example in Unreal Engine 5 there is a system called 'Nanite' for rendering large objects, like landscapes, by approximating them with so many tiny triangles that each triangle takes up close to a single pixel on the screen.



Nanite tessellating a rocky surface, from https://dev.epicgames.com/documentation/en-us/unreal-engine/nanite-virtualized-geometry-in-unreal-engine. On the right, each triangle has been given a different colour.

To display the landscape we need to know how much sunlight bounces off the tiny triangular fragment. This depends on the angle (both the angle to the sun, so we know how much light hits it, and the angle to the camera so we know how much is reflected towards our viewpoint) and the orientation (a bit of rock facing away from the sun will look very different to one facing towards it!), and also on the surface area which determines how much light the tiny polygon can reflect. However, because the polygon is so small, its shape doesn't matter for calculating the flux of light, so it behaves like a vector from our strange vector space! We discuss how this space can be constructed mathematically from ordinary 3D Euclidean space in a subsequent section.

Interesting mathematical examples of vector spaces

There are also some examples which show that the notion of a real vector space is more general than you might at first have thought which we *can* define using only the machinery studied in this unit:

Example 9.11. We state above that the real numbers $\mathbb{R}$ form a vector space over the reals for the usual addition and multiplication. However it is important to note that was can *scale* the scalar multiplication by any factor and get another vector space over the reals. For example, let's decide to scale all our multiplications by a factor of 2. Let us write $\underline{(r)}$ for the real number r thought of as an element of this funny vector space. Then we have

$$s\underline{(r)} = \underline{(2sr)}$$

You should check that all the properties of a vector space are still satisfied!

In a sense this is "the same" vector space as $\mathbb{R}$, because there is a unique sensible way of converting between them (we explore this more in the next section) – but in a more

concrete sense it is clearly a bit different!

Example 9.12. The set $\mathbb{R}^+ = \{x \in \mathbb{R} | 0 < x\}$ is a vector space when equipped with the following operations, where we write $\underline{(x)}$ for the element $x \in \mathbb{R}^+$ thought of as a vector. Note that the zero vector is not $\underline{(0)}$, which might be a surprise!

$$\underline{(v)} + \underline{(w)} = \underline{(vw)}$$

$$r\underline{(v)} = \underline{(v^r)}$$

$$\underline{0} = \underline{(1)}$$

CExercise 201. (on Sheet 18) Show that the vector space defined in Example 9.12 satisfies all the properties of a vector space.

Example 9.13. Suppose we have a set of hypotheses $\{h_1, ...h_n\}$ which we consider possible. We are interested in the set of probability distributions on $\{h_1, ...h_n\}$ which do not send any element to 0 (since we consider them all to be possible). This set of probability distributions forms a vector space under the following operations, where given a probability distribution $\underline{v}$ we will write $P_{\underline{v}}(h_i)$ to mean the probability of the hypothesis h_i according to the probability distribution $\underline{v}$

Then we define the vector sum of probability distributions by

$$P_{\underline{v}+\underline{w}}(h_i) = \frac{P_{\underline{v}}(h_i)P_{\underline{w}}(h_i)}{\sum_{k=1}^{n} P_{\underline{v}}(h_k)P_{\underline{w}}(h_k)}$$

and the scalar product by

$$P_{r\underline{v}}(h_i) = \frac{P_{\underline{v}}(h_i)^r}{\sum_{k=1}^{n} P_{\underline{v}}^r}$$

This gives a vector space structure.

Exercise 202. Show that we can make the set $\{\emptyset\}$, i.e. the set containing the empty set, into a vector space over the real numbers. Make sure you define vector addition, scalar multiplication, and the zero vector, and prove that your proposed definitions satisfy all the properties required by the definition.

Is it possible to do the same for the empty set $\emptyset$?

9.1.2 Fields and general vector spaces

From an algebraic point of view, only certain properties of the real numbers are used when studying vector spaces. This means we can consider vector spaces whose scalars come from other sets, as long as they share these important properties, expanding the number of applications of linear algebra. We call a set with a suitable structure to be a set of scalars of a vector space a 'field'.

Definition 69: field

A **field** is a set K with two binary operations $+ : K \times K \to K$ and $\cdot : K \times K \to K$ satisfying the following properties:

- The operation $+$ is commutative, associative, has a unit 0 and every element $r \in K$ has an additive inverse $-r$.
- The oparation $\cdot$ is commutative, associative, and has a unit 1
- For all $r \in K$, if $r \neq 0$ then there is an element r^{-1} which is a multiplicative inverse for r (that is, an inverse for the operation $\cdot$)
- Multiplication distributes over addition, that is for all $a, b, c \in K$ we have $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$

Usually we write $a \cdot b$ simply as ab, and we make use of all the usual bracketing conventions.

Example 9.14. The rational numbers $\mathbb{Q}$, real numbers $\mathbb{R}$, and complex numbers $\mathbb{C}$ are all fields with the usual operations.

Example 9.15. The set $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is a field when equipped with the restriction of the usual addition and multiplication from $\mathbb{R}$. First, we have to check that the does successfully define binary operations on $\mathbb{Q}[\sqrt{2}]$, in other words that the sum and product of elements of $\mathbb{Q}[\sqrt{2}]$ are again elements.

By calculations in $\mathbb{R}$ we see that

- $(a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + b) + (c + d)\sqrt{2}$ and
- $(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$,

which are elements of $\mathbb{Q}[\sqrt{2}]$ as required.

Now associativity and commutativity of the operations, as well as distributivity, follow from these properties for $\mathbb{R}$. (If you have a go at proving them, you find each one is immediate by a "calculation in $\mathbb{R}$").

We need to check that the additive and multiplicative inverses of elements are again elements of $\mathbb{Q}[\sqrt{2}]$.

- We have $(a + b\sqrt{2}) + (-a + -b\sqrt{2}) = 0$ and the latter is an element of $\mathbb{Q}\sqrt{2}$ and
- if $a \neq 0$ or $b \neq 0$ then we need to find a multiplicative inverse for $a + b\sqrt{2}$. By calculations in $\mathbb{R}$ we know that if this inverse exists it must be of the form

$$\frac{a}{a^2 - 2b^2} + \frac{b}{a^2 - 2b^2}\sqrt{2}$$

  but of course this expression may not be well-defined if $a^2 - 2b^2 = 0$ (thinking about fields can help us to remember that we only have multiplicative inverses for non-zero elements!). If we assume $b \neq 0$ then $a^2 - 2b^2 = 0$ implies that $a/b = \sqrt{2}$, but this can't happen since $a/b \in \mathbb{Q}$ but $\sqrt{2}$ is irrational. If $b = 0$ then $a^2 - 2b^2 = 0$ implies

$a = 0$, but this contradicts the assumption we started with—if they are both zero then this is the zero element of $\mathbb{Q}[\sqrt{2}]$ and we do not need to supply a multiplicative inverse for *that*!

Example 9.16. (Non-example!) The natural numbers $\mathbb{N}$ are not a field – for example, they lack additive inverses. Similarly, the integers $\mathbb{Z}$ are not a field, because there are non-zero integers which do not have multiplicative inverses.

Example 9.17. In an exercise in a previous section, we saw that $\mathbb{Z}_p$, namely the set $\{0, ..., (p-1)\}$ with addition and multiplication defined by adding and multiplying in $\mathbb{Z}$ then doing $\mod p$, is a field if and only if p is a prime number. In that case every element except zero has a multiplicative inverse as proved in a previous exercise.

The following exercise is a good start for those interested fields mathematically, or in how finite fields are applied in cryptography.

Exercise 203. Show that the set $\{0, 1\} \times \{0, 1\}$ is a field when equipped with the following operations:

- $(a, b) + (c, d) = (a + b \mod 2, c + d \mod 2)$
- $(a, b) \cdot (c, d) = (ac + bd \mod 2, ad + bc \mod 2)$

Is it the case that in every field, if an element r satisfies the equation $r^2 = 1$ then we have either $r = 1$ or $r = -1$?

Exercise 204. Show that $\mathbb{Q} \times \mathbb{Q}$ is ***not*** a field when equipped with the following operations:

- $(a, b) + (c, d) = (a + b, c + d)$
- $(a, b) \cdot (c, d) = (ac, ad + bc)$

(It does however have almost all the properties – mathematicians would say it is a "ring". In particular evaluating polynomials is familiar in this structure. You may find it amusing to evaluate $P((x, 1))$ for your favourite polynomial P)

Given any field, the definition of a vector space is almost identical to that for the the reals. Nevertheless, we state it here in full, since this is the "official" definition of vector space, for which a "vector space over the reals" as defined in the previous section forms only one kind of example.

Definition 70: Vector space

Let F be any field. A **vector space over** F (often shortened to "F-vector space") is a set V together with

- A binary operation $+ : V \times V \to V$ called "vector addition",
- An element $\underline{0} \in V$
- A function from $F \times V$ to V called "scalar multiplication". For $r \in F$ and $\underline{v} \in V$ we

simply write $r\underline{v}$ to mean the scalar multiplication of r and $\underline{v}$

satisfying the following properties:

- $+$ is commutative and associative
- $\underline{0}$ is a unit for $+$
- every element $\underline{v}$ of V has an inverse for the operation $+$ which we write as $-\underline{v}$

And for all $r, s \in F$ and $\underline{v}, \underline{w} \in V$ we have

- $1\underline{v} = \underline{v}$, we say that 1 is a **scalar unit**
- $r(s\underline{v}) = (rs)\underline{v}$, which we call **scalar compatibility**
- $(r + s)\underline{v} = r\underline{v} + s\underline{v}$, which we call **scalar dsitributivity**
- $r(\underline{v} + \underline{w}) = r\underline{v} + r\underline{w}$, which we call **vector distributivity**

Whenever one describes a vector space, one must first say over which field one is working. However, in these notes since we often want to describe properties of general vector spaces over general fields, we often leave this implicit, and simply use the word **scalar** to refer to an element of the underlying field.

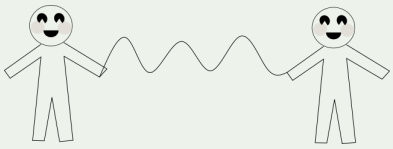
Example 9.18. Every field F is a vector space over itself when we consider its own addition and multiplication operations. We use the name of the field K to refer to this vector space. (This is somewhat ambigious since as we have seen there are vector spaces with the same underlying set of vectors, but "weird" addition or multiplication operations. However, we use the convention that we have to find other names for these, and the vector space we call F is always the one with the "usual" operations inherited from F).

EExercise 205. (on Sheet 18) Let X be a set. Show that the powerset $\mathcal{P}X$ is a vector space over the field $\mathbb{Z}_2$ when equipped with the following operations:

- For all $A \subseteq X$ we define $0A = \emptyset$
- For all $A \subseteq X$ we define $1A = A$
- For all $A, B \subseteq X$ we define $A + B = \{x \in X \mid x \in A \cup B \text{ and } x \notin A \cap B\}$

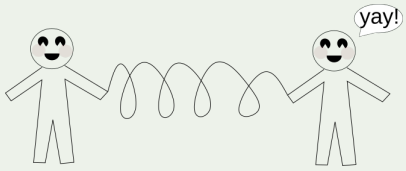
Vector spaces over the complex numbers are highly important to modelling physical processes involving wave phenomena. This usually arises because waves of the same frequency can have different amplitudes (e.g. how loud a note sounds), but also different *phases*: how far through the cycle the wave is at a given point. Because phase is a position in a cyclic process, it can be thought of as a point on a circle. It turns out that in many applications, thinking of the amplitude and phase of a wave as the absolute and argument of a complex number makes the addition and multiplication of complex numbers meaningful for computing interactions of these waves. This gives rise to various examples which we point out conceptually, but we shy away from the mathematical details as these lead us away from linear algebra.

Example 9.19. (Classical waves on a string, polarization of light) Suppose you and your friend are entertaining yourselves by sending waves down a ribbon:



It is clear that it is possible to waggle the ribbon in more ways that just up and down. You could also send waves by waggling the ribbon in the horizontal direction. But you could also waggle the ribbon in a diagonal direction. It is not too difficult conceptually to think of the diagonal waggling as being "both horizontal and vertical waggling at the same time". We can think of these kinds of wagglings as forming a vector space over the reals, in which waggling at different angles is equal to the sum of horizontal and vertical waggling, scaled by various amounts which determine the angle.

But it is also possible to move your hand in a circle, sending loopy waves down the ribbon:



It may be harder to think of this kind of motion as being horizontal and vertical waggling happening at the same time. But your hand *is* waggling horizontally and vertically, just no longer synchronized so that the maximum horizontal displacement happens at the same moment as the maximum vertical displacement – in other words the horizontal and vertical wagglings have gone *out of phase*. Physicists model this type of scenario using a complex vector space, in which we may have equations like

$$\text{“circular waggling”} = \frac{1}{\sqrt{2}}\text{“horizontal waggling”} + \frac{i}{\sqrt{2}}\text{“vertical waggling”}$$

with details of signs and where the i goes depending on issues like whether the circular motion is clockwise or anticlockwise, etc.

Roughly speaking the $1/\sqrt{2}$ is not very exciting and just takes care of the fact that we want "the same amount" of waggling, not more when we add the two kinds of waggling together, and the same factor would appear if we were describing diagonal waggling. The interesting thing is the factor of i which shows we need wagglings to form a complex vector space. Intuitively, since multiplying by i rotates a complex number by a quarter of a circle, we are accounting for the fact that the maximum vertical displacement now happens a quarter of a cycle out from the maximum horizontal displacement, which is what is needed to make a circular motion.

The same ideas apply to light waves. One can think of a light beam as moving in a particular direction, and free to oscillate in the space of directions perpendicular to its motion, like a tiny version of the ribbon. Thus it can exhibit polarization phenomena which are mathematically equivalent to those above. (In fact, a real ribbon can twist, stretch, and do lots of other complicated things, so the discussion really applies to the simpler case of light much better). This helps explain puzling facts about optics, such as the fact that a linearly polarized filter does not completely block circularly polarized light.

Example 9.20. (Quantum computing) One of the postulates of quantum mechanics is that if a quantum system can be in certain states, then it can also be in a state, called a "superposition" of those states, which can be thought of as a combination of those states in various amounts. However in order to reproduce the ways in which interference happens in quantum systems, these "amounts" have to be allowed to be complex numbers. This notion of combination obeys all the properties of vector addition, and increasing the "amount" by a complex number obeys the properties of scalar multiplication, so another way to describe this postulate is that states of a system form a complex vector space.

For example, if we want to have a memory location in a quantum computer capable of holding two bits, it must be possible to measure it as being in at least the four usual states of a two bit system, which physicists write as $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. This means it can also be in any of the states

$$a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$

for complex numbers a, b, c, and d. (Technically not all of these "state vectors" are physically possible states, but they are used in calculations, and the physics is beyond what we are interested in here.)

An important way of generating new examples of vector spaces out of old ones is by generalizing the idea of Example 9.1.

Definition 71: direct sum of vector spaces

Let K be a field and V and W be vector spaces over K. Then we define the vector space $V \oplus W$ over K as follows:

- The vectors of $V \oplus W$ are the elements of the set $V \times W$
- We define $(a, b) + (c, d) = (a + c, b + d)$ for all $a, c \in V$ and $b, d \in W$
- We define $r(a, b) = (ra, rb)$ for all $r \in F$, $a \in V$ and $b \in W$

Note that in the above $+$ has three meanings. We are defining the meaning of $+$ for vectors in $V \oplus W$ by using the operation $+$ defined on V in the left component of the pair, and the (usually different!) operation $+$ define on W in the right component. You may be confused as to why we don't simply call this vector space $V \times W$ but, as we have seen, there are other ways of defining operations on the set of pairs which make it into a vector space, so it is useful to have an unambiguous name for the simplest way of doing this.

However, we do adopt the convention that when we write V^n, we mean $V \oplus \ldots \oplus V$ where we have used n copies of V.

Exercise 206. For all fields K and K-vector spaces V and W, show that $V \oplus W$ is indeed a vector space over K.

9.1.3 Subspaces, linear independence, and bases

Definition 72: linear combination

Given vectors $\underline{v_1}, \ldots, \underline{v_n}$ in a vectors space V a **linear combination** of $\underline{v_1}, \ldots, \underline{v_n}$ is any vector of the form $r_1\underline{v_1} + \ldots + r_n\underline{v_n}$.

A linear combination of vectors is any sum of scaled versions of the vectors. To be more precise, we could define this '$+ \ldots +$' notation by recursion on n:

- **Base case**: if $n = 0$ then the linear combination is the zero vector $\underline{0}$.
- **Step case**: by the notation $r_1\underline{v_1} + \ldots + r_{n+1}\underline{v_{n+1}}$ we mean $(r_1\underline{v_1} + \ldots + r_n\underline{v_n}) + r_{n+1}\underline{v_{n+1}}$

Example 9.21. Each of the vectors $\underline{v_1}, \ldots, \underline{v_n}$ by itself if a linear combination of the vectors $\underline{v_1}, \ldots, \underline{v_n}$. For example if we wanted to express $\underline{v_1}$ as a linear combination of $\underline{v_1}, \ldots, \underline{v_n}$ we would choose $r_1 = 1$ and $r_i = 0$ when $i > 1$. Then we have $1\underline{v_1} + \ldots + 0\underline{v_n} = \underline{v_1}$ by using Theorem 9.1.

Exercise 207. Show that for any set of vectors $\underline{v_1}, \ldots, \underline{v_n}$, the zero vector $\underline{0}$ is a linear combination of $\underline{v_1}, \ldots, \underline{v_n}$.

The set of linear combinations of a fixed set of vectors is closed under the operations of the vectors space:

Proposition 9.2

Given vectors $\underline{v_1}, \ldots, \underline{v_n}$ in a vectors space V and linear combinations $w_1 = r_1\underline{v_1} + \ldots + r_n\underline{v_n}$ and $w_2 = s_1\underline{v_1} + \ldots + s_n\underline{v_n}$, then the sum $w_1 + w_2$ is the linear combination $(r_1 + s_1)\underline{v_1} + \ldots + (r_n + s_n)\underline{v_n}$.

Proof. By induction on n.
Base case: If $n = 1$ then $\underline{w_1} = r_1\underline{v_1}$ and $\underline{w_2} = s_1\underline{v+1}$. Hence $w_1 + w_2 = r_1\underline{v_1} + s_1\underline{v+1}(r_1 + s_1)\underline{1}$ as required.

Step case: If $w_1 = r_1\underline{v_1} + \ldots + r_n\underline{v_n} + r_{n+1}\underline{v_{n+1}}$ and $w_2 = s_1\underline{v_1} + \ldots + s_n\underline{v_n} + s_{n+1}\underline{v_{n+1}}$ then we have

$$
\begin{aligned}
&\underline{w_1} + \underline{w_2} \\
=&(r_1\underline{v_1} + \ldots + r_n\underline{v_n} + r_{n+1}\underline{v_{n+1}}) + (s_1\underline{v_1} + \ldots + s_n\underline{v_n} + s_{n+1}\underline{v_{n+1}}) && \text{def. } w_1, w_2 \\
=&(r_1\underline{v_1} + \ldots + r_n\underline{v_n} + s_1\underline{v_1} + \ldots + s_n\underline{v_n}) + r_{n+1}\underline{v_{n+1}} + s_{n+1}\underline{v_{n+1}} && \text{assoc. comm. } + \\
=&((r_1 + s_1)\underline{v_1} + \ldots + (r_n + s_n)\underline{v_n}) + r_{n+1}\underline{v_{n+1}} + s_{n+1}\underline{v_{n+1}} && \text{by I.H.} \\
=&((r_1 + s_1)\underline{v_1} + \ldots + (r_n + s_n)\underline{v_n}) + (r_{n+1} + s_{n+1})\underline{v_{n+1}} && \text{distributivity} \\
=&(r_1 + s_1)\underline{v_1} + \ldots + (r_n + s_n)\underline{v_n} + (r_{n+1} + s_{n+1})\underline{v_{n+1}} && \text{assoc. } +
\end{aligned}
$$

as required.

Proposition 9.3

Given vectors $\underline{v_1}, \ldots, \underline{v_n}$ in a vector space V and a linear combination $w = r_1\underline{v_1} + \ldots + r_n\underline{v_n}$ then the scalar product $s\underline{w}$ is the linear combination $(sr_1)\underline{v_1} + \ldots + (sr_n)\underline{v_n}$.

Proof. Exercise.

Now we have seen what happens when we hold a set of vectors fixed and ask whether various vectors are linear combinations of them, we are also interested in the reverse question: in what ways can we vary a set of vectors such that their linear combinations remain the same.

Proposition 9.4

Let V be a vector space and $\underline{v_1}, \underline{v_2}, \ldots, \underline{V_n}$ be vectors in v. Suppose $\underline{w}$ is a linear combination of $\underline{v_1}, \underline{v_2}, \ldots, \underline{V_n}$. Then $\underline{w}$ is also a linear combination of $(r\underline{v_1} + s\underline{v_2}), \underline{v_2}, \ldots, \underline{v_n}$ for all scalars r and s such that $r \neq 0$.

Proof. Since w is a linear combination of $\underline{v_1}, \underline{v_2}, \ldots, \underline{V_n}$, we can find scalars such that $\underline{w} = r_1\underline{v_1} + r_2\underline{v_2} + \ldots + r_n\underline{v_n}$.

Then

$$\begin{aligned}
\underline{w} &= r_1\underline{v_1} + r_2\underline{v_2} + \ldots + r_n\underline{v_n} \\
&= (r_1/r)r\underline{v_1} + ((sr_1)/r - (sr_1)/r + r_2)\underline{v_2} + \ldots + r_n\underline{v_n} && \text{calc. with scalars} \\
&= (r_1/r)r\underline{v_1} + ((sr_1)/r)\underline{v_2} + (-(sr_1)/r + r_2)\underline{v_2} + \ldots + r_n\underline{v_n} && \text{scalar distr.} \\
&= (r_1/r)(r\underline{v_1} + s\underline{v_2}) + (-(sr_1)/r + r_2)\underline{v_2} + \ldots + r_n\underline{v_n} && \text{vector distr.}
\end{aligned}$$

which is a linear combination of $(r\underline{v_1} + s\underline{v_2}), \underline{v_2}, \ldots, \underline{v_n}$ as required.

This principle is the basis of most "school" methods of solving linear equations, such as "Gaussian elimination". Many basic tasks in linear algebra come down to solving systems of linear equations; you should feel free to do this however you like. The intended focus of this section of the unit is on the conceptual aspects of how the various ideas of linear algebra fit together rather than on calculations. And modern algorithms, at least for vector spaces over $\mathbb{R}$ and $\mathbb{C}$, use very sophisticated methods which are faster and less error prone than the working out we do by hand.

Definition 73: subspace

Let K be a field and V be a vector space over K. Then a subset $W \subseteq V$ is called a **subspace** if and only if we have:

- $\underline{0} \in W$
- For all $r \in K$ and $\underline{w} \in W$ we have $r\underline{w} \in W$
- For all $\underline{w}, \underline{w'} \in W$ we have $\underline{w} + \underline{w'} \in W$

Note that by Theorem 9.1, if we check the second two properties, then the first is guaranteed to hold.

Example 9.22. In any vector space V, the whole space V is always a subspace, this is immediate from the definition. It it a little more work to see that $\{\underline{0}\}$ us a subspace. Closure under vector addition holds because $\underline{0}$ is an additive unit. Closure under scalar multiplication holds by Theorem 9.1.

Example 9.23. In the 2D Euclidean vector space, if we pick a direction in space, then the set of all vectors pointing in that direction (or in the opposite direction) form a subspace. As described above, the whole space and the set just containing the zero vector are subspaces, and these are all subspaces.

These idea continue to give examples in 3D, but we have more subspaces: for each plane the set of all vectors lying parallel to the plane is a subspace of the 3D Euclidean vector space.

Exercise 208. Let K be a field. Consider the vector space of functions from a set X to a field K. In the case $K = \mathbb{R}$ this is defined in Exercise 200, and the argument work analogously over any field.
Say that a function f has 'finite support' if and only if the set $\{x \in X \mid f(x) \neq 0\}$ is finite. Show that the set of functions with finite support is a subspace of the vector space of functions.

Definition 74: linear independence

Let V be a vector space and $S \subseteq V$ be a set of vectors. We say that S is **linearly independent** if and only if for all finite subsets $\{\underline{v_1}, ...\underline{v_n}\} \subseteq S$, if there are scalars $r_1, ...r_n$ such that $r_1\underline{v_1} + ... + r_n\underline{v_n} = \underline{0}$ then for all $1 \leq i \leq n$ we have $r_i = 0$.

In other words no combination of vectors from S can be made to "cancel each other out" and get the zero vector unless this is achieved trivially by scaling them all by 0.

Definition 75: span

Let V be a vector space and $S \subseteq V$ be a set of vectors in V. The set

$$\mathbf{span}(S) = \{r_1\underline{v_1} + ... + r_n\underline{v_n} \mid \{\underline{v_1}, ..., \underline{v_n}\} \subseteq S \text{ and } r_1, ..., r_n \text{ are scalars}\}$$

Proposition 9.5

Let V be a vector space and $S \subseteq V$ be a set of vectors in V. Then $\mathbf{span}(S)$ is a subspace of V

Proof. First we show that $\mathbf{span}(S)$ is closed under scalar multiplication. Given a vector $r_1\underline{v_1} + ... + r_n\underline{v_n}$ ins $\mathbf{span}(S)$ and a scalar s, we have

$$\begin{aligned} &s(r_1\underline{v_1} + ... + r_n\underline{v_n}) \\ =&s(r_1\underline{v_1}) + ... + s(r_n\underline{v_n}) \\ =&(sr_1)\underline{v_1} + ... + (sr_n)\underline{v_n} \end{aligned}$$

by using Proposition 9.3, and this is an element of $\mathbf{span}(S)$ by definition.
Now we must show that $\mathbf{span}(S)$ is closed under addition of vectors. To do so, suppose we have two vectors $\underline{v} = r_1\underline{v_1} + ... + r_n\underline{v_m}$ and $\underline{w} = s_1\underline{w_1} + ... + s_n\underline{w_n}$ in $\mathbf{span}(S)$. Now consider the set of vectors $\{u_1, ..., u_k\} = \{v_1, ...v_m\} \cup \{w_1, ...w_n\}$. (Note that k is not necessarily equal to $m + n$ because the two sets may have some vectors in common, and that this is again a subset of S as it is the union of two subsets of S).

Then we can find scalars $r'_1, ...r'_k$ such that $\underline{v} = r'_1\underline{u_1} + ... + r'_k\underline{u_k}$ by setting r'_i to r_j if $u_i = v_j$ and to 0 if no such j can be found. Similarly, we can find $s'_i, ...s'_k$ such that

$\underline{w} = s'_1\underline{u_1} + \ldots + s'_k\underline{u_k}$. Thus by Proposition 9.2

$$\underline{v} + \underline{w} = (r'_1 + s'_1)\underline{u_1} + \ldots + (r'_k + s'_k)\underline{u_k}$$

and hence is an element of $\mathbf{span}(S)$ as required.

We say that a subspace W is "spanned by" S, or that S "spans" W if $\mathbf{span}(S) = W$.

Definition 76: basis

A list of vectors $\underline{v_1}, \ldots, \underline{v_n}$ in a vector space V is a **basis** of V if and only if the set $\{\underline{v_1}, \ldots, \underline{v_n}\}$ is linearly independent and spans V.

Note that while most of our definitions are made for sets of vectors, here I have defined a basis to be a list. In some books they define a basis to be a set, similar to the other notions, and call the notion defined above an "ordered basis". As we shall see, the notion of a basis gives us a very concrete description of the vectors in a vector space, in particular a way to write them down unambiguously. But we write things down in a particular order, so once we start doing this we are effectively working with the ordered notion of basis. However for more abstract vector spaces there may be no obvious order, and in those circumstances we will use the term **unordered basis** to mean a set of vectors satisfying the conditions above, rather than a list.

Theorem 9.6

Every vector space has a basis, and all bases for the same vector space have the same size as sets.

Proving this theorem for infinite-dimensional vector spaces involves set theory which goes well beyond this course unit; on the other hand a finite-dimensional vector space is defined to be one with a finite basis set, so the existence of a basis is not very exciting.

Definition 77: column vector notation

Let V be a vector space and $B = \underline{b_1}, \ldots \underline{b_n}$ be a basis of V. Then we write

$$\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}_B = c_1\underline{b_1} + \ldots + c_n\underline{b_n}$$

Note that since B spans V by definition, every vector can be written in this form. Using the fact that B is linearly independent, we can show that this representation is *unique*, which means it is easy to see when two vectors in a vector space are equal

The use of a basis also allows us to calculate sums and scalings of vectors easily:

Theorem 9.7

Let V be a vector space and $B = \underline{b_1}, \ldots \underline{b_n}$ be a basis of V. Then

$$\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}_B + \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}_B = \begin{pmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{pmatrix}_B$$

and

$$s\begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}_B = \begin{pmatrix} sc_1 \\ \vdots \\ sc_n \end{pmatrix}_B$$

Proof. See exercise below.

CExercise 209. (on Sheet 18) Prove Theorem 9.7

Exercise 210. Consider a set X and a field K. The functions of finite support from X to K form a vector space; for $K = \mathbb{R}$ you proved this in Exercise 208, and you may wish to check that your argument only uses the fact that $\mathbb{R}$ is a field. Thus they form a vector space for any field K. For all $x \in X$ define $\bar{x}$ to be the function which sends x to 1 and all other elements of X to 0. Show that the set $\{\bar{x} \mid x \in X\}$ forms an unordered basis for the space of functions of finite support.

9.1.4 Quotients of vector spaces

In the previous part of this unit, we look at equivalence relations and their quotient sets. For vector spaces, certain equivalence relations are well behaved.

Definition 78: equivalence corresponding to a subspace

Let V be a vector space and W be a subspace of V. Then we define a relation $\sim_W$ on V which we call the **equivalence relation corresponding to** W which is defined by stipulating that, for all $\underline{v_1}, \underline{v_2} \in V$ we have $\underline{v_1} \sim_W \underline{v_2}$ if and only if there is a vector $\underline{w} \in W$ such that $v_2 = v_1 + w$

Exercise 211. Prove that for all vector spaces V and subspaces W the relation $\sim_W$ defined above is an equivalence relation.

The interesting thing about equivalence relations of this form is that the quotient set can be given the structure of a vector space, based on the structure of the original vector space.

Theorem 9.8

Let V be a vector space and W be a subspace of V. Then the quotient set $V/\sim_W$ is a vector space when we define the operations as follows.

For all vectors $\underline{v}$ and $\underline{v'}$ and scalars r we define

- $[\underline{v}] + [\underline{v'}] = [\underline{v} + \underline{v'}]$
- $r[\underline{v}] = [r\underline{v}]$

Proof. The first thing we have to do is make sure the definitions of vector addition and scalar multiplication make sense, i.e. that no matter which representatives of the input equivalence classes we pick, the outputs are all elements of the same equivalence class.

For vector addition, let $\underline{v_1} \sim_W \underline{v_2}$ and $\underline{v_1'} \sim_W \underline{v_2'}$. Then we need to show that $v_1 + v_1' \sim_W v_2 + v_2'$. Now since $\underline{v_1} \sim_W \underline{v_2}$ there is a $\underline{w} \in W$ such that $\underline{v_2} = \underline{v_1} + \underline{w}$ and since $\underline{v_1'} \sim_W \underline{v_2'}$

there is a $\underline{w'} \in W$ such that $\underline{v_2'} = \underline{v_1'} + \underline{w'}$ now we have

$$\begin{aligned}
&\underline{v_2} + \underline{v_2'} \\
=&\underline{v_1} + \underline{w} + \underline{v_1'} + \underline{w'} && \text{assumption} \\
=&\underline{v_1} + \underline{v_1'} + (\underline{w} + \underline{w'}) && \text{assoc. comm. } +
\end{aligned}$$

Now since W is a subspace, we have $\underline{w} + \underline{w'} \in W$, and so the above equality shows that $\underline{v_1} + \underline{v_1'} \sim_W \underline{v_2} + \underline{v_2'}$ as required.

For scalar multiplication, let $\underline{v_1} \sim_W \underline{v_2}$, which means that there exists a $w \in W$ such that $\underline{v_2} = \underline{v_1} + \underline{w}$. Then

$$\begin{aligned}
&r\underline{v_2} \\
=&r(\underline{v_1} + \underline{w}) && \text{assumption} \\
=&r\underline{v_1} + r\underline{w} && \text{vector distributivity}
\end{aligned}$$

Now since W is a subspace, we have $r\underline{w} \in W$, which means that $r\underline{v_1} \sim_W r\underline{v_2}$ as required.

Now we have proved that the operations make sense, we will have to check all the properties of a vector space. However, these become a simple matter of expanding the definition and using the properties of the original vector space V. For example, we prove that vector distributivity holds as follows.

$$\begin{aligned}
&r([\underline{v}] + [\underline{v'}]) \\
=&r([\underline{v} + \underline{v'}]) && \text{def. } + \\
=&[r(\underline{v} + \underline{v'})] && \text{def. scalar mult.} \\
=&[r\underline{v} + r\underline{v'}] && \text{vector distr. in } V \\
=&[r\underline{v}] + [r\underline{v'}] && \text{def. } + \\
=&r[\underline{v}] + r[\underline{v'}] && \text{def. scalar mult}
\end{aligned}$$

A similar argument establishes the remaining properties.

One of the most important constructions which produces a vector space given two input vector spaces can be defined using quotients, although the resulting definition is rather hard to understand. In a subsequent section, we explore an alternative definition which is more intuitive. Nevertheless, it is useful to have seen the definition below, since despite looking a bit scary, it has quite a straightforward meaning in practice.

Definition 79: tensor product

Given vector spaces V and W over a field K, the **tensor product** $V \otimes W$ of V and W is defined as follows. First, recall from Exercise 208 and Exercise 210 that for any set X that the set of functions with finite support from X to K form a K-vector space. Let L be the set of functions with finite support from the set $V \times W$ to K.

Using the notation from Exercise 210, we define the following set of vectors:

$$\begin{aligned}
R =& \{\overline{(\underline{v_1} + \underline{v_2}, \underline{w})} - \overline{(\underline{v_1}, \underline{w})} - \overline{(\underline{v_2}, w)} \mid \underline{v_1}, \underline{v_2} \in V, \underline{w} \in W\} \\
&\cup \{\overline{(\underline{v}, \underline{w_1} + \underline{w_2})} - \overline{(\underline{v}, \underline{w_1})} - \overline{(\underline{v}, \underline{w_2})} \mid \underline{v} \in V, \underline{w_1}, \underline{w_2} \in W\} \\
&\cup \{\overline{(s\underline{v}, \underline{w})} - \overline{s(\underline{v}, \underline{w})} \mid \underline{v} \in V, \underline{w} \in W, s \in K\}
\end{aligned}$$

$$\cup \{\overline{(\underline{v}, s\underline{w})} - \overline{s(\underline{v}, \underline{w})} \mid \underline{v} \in V, \underline{w} \in W, s \in K\}$$

Now we define $V \otimes W = L/\sim_{\mathbf{span}(R)}$

As promised this looks rather daunting. What is really going on here is that we are defining a vector space in which certain algebraic laws are guaranteed to hold. For example, because we have put all elements of the form $\overline{(\underline{v_1} + \underline{v_2}, \underline{w})} - \overline{(\underline{v_1}, \underline{w})} - \overline{(\underline{v_2}, w)}$ in the subspace $\mathbf{span}(R)$ used to define the equivalence relation, we are demanding that in the quotient space the equivalence class of this element is the same as $[0]$, or in other words we are demanding that

$$[\overline{(\underline{v_1} + \underline{v_2}, \underline{w})}] = [\overline{(\underline{v_1}, \underline{w})}] + [\overline{(\underline{v_2}, w)}].$$

All the overlines and underlines become messy, so let's use the traditional notation for elements of $V \otimes W$ where we denote $[\overline{(\underline{v}, \underline{w})}]$ by $\underline{v} \otimes \underline{w}$. Then the definition of $V \otimes W$ says that in this space we are guaranteed to have the equations

$$\begin{aligned}
(\underline{v_1} + \underline{v_2}) \otimes w &= \underline{v_1} \otimes \underline{w} + \underline{v_2} \otimes \underline{w} \\
\underline{v} \otimes (\underline{w_1} + \underline{w_2}) &= \underline{v} \otimes \underline{w_1} + \underline{v} \otimes \underline{w_2} \\
(s\underline{v}) \otimes \underline{w} &= s(\underline{v} \otimes \underline{w}) \\
\underline{v} \otimes (s\underline{w}) &= s(\underline{v} \otimes \underline{w})
\end{aligned}$$

wherever they make sense.

What is good about this? It shows how much flexibility we have in constructing vector spaces. For one motivation, recall that the state spaces of quantum systems form complex vector spaces. Suppose we have two systems with state spaces V and W. What happens if we put these two systems together to make a bigger physical system? Physicists found that there is an operation which combines states $\underline{v} \in V$ and $\underline{w} \in W$ to form a state $\underline{v} \otimes \underline{w}$ of the combined system. However, this operation was found to obey the equations given above. One might have thought that you'd need to search for decades to find a vector space with an operation satisfying these algebraic laws. However it turns out that Theorem 9.8 which seemed relatively easy to prove has done the job for us! The only "clever" bit is the idea of using functions of finite support on the product set $V \times W$, but this is the standard trick to use when you want an operation turning pairs in this set into elements of your desired vector space. Of course, this vector space is rather mysterious when defined in this way. For example, what is its dimension? Do the equations above imply that every vector is equal to $\underline{0}$? The answers to these are not at all obvious! However, what the construction has done is turned this from a modelling problem about the real world into a purely mathematical question about this abstract, but unambiguously specified, vector space.

A further quotient gives rise to a vector space which is more concrete and commonly used in engineering and computer graphics – the vector space of little polygons from Example 9.10! Suppose we think for a while about the real vector space $\mathbb{R}^3 \otimes \mathbb{R}^3$. We know that for vectors $\underline{v}, \underline{w} \in \mathbb{R}^3$ there is a vector $\underline{v} \otimes \underline{w} \in \mathbb{R}^3 \otimes \mathbb{R}^3$ and we could try to think of this as the little parallelogram such that one side is $\underline{v}$ and the other side is $\underline{w}$. This geometric interpretation certainly satisfies the equations for the $\otimes$ operation listed above. For example, if we stretch one side of a parallelogram by a factor s then we get a parallelogram with the same orientation in 3D space, but with an areas s times larger. Since in our intended vector space we care about the orientation and area, but not the shape, we don't care which side was stretched, so it makes sense to have $(s\underline{v}) \otimes \underline{w} = s(\underline{v} \otimes \underline{w}) = \underline{v} \otimes (s\underline{w})$ as the last two equations above tell us.

However, there is an obvious geometric fact which is not included in those equations. If we take *the same vector twice*, then the parallelogram formed with those sides has zero area. To account for this fact, we need to make a further quotient.

Definition 80: exterior square

Given a vector space V the **exterior square** $V \wedge V$ is defined as follows.

First consider the tensor product $V \otimes V$. Within this we have a subspace

$$D = \mathbf{span}(\{\underline{v} \otimes \underline{v} \mid \underline{v} \in V\})$$

. Then we define

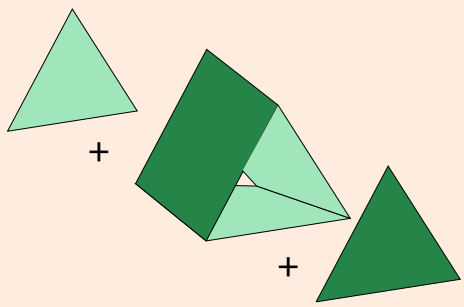
$$V \wedge V = V \otimes V / \sim_D$$

Following tradition, we write $\underline{v} \wedge \underline{w}$ for the equivalence class $[\underline{v} \otimes \underline{w}]$. Then the definition of $V \wedge V$ means we have the following equations, should we want them:

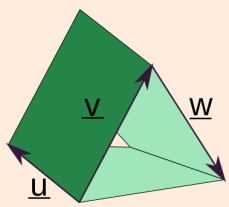
$$\begin{aligned}
(\underline{v_1} + \underline{v_2}) \wedge w &= \underline{v_1} \wedge \underline{w} + \underline{v_2} \wedge \underline{w} \\
\underline{v} \wedge (\underline{w_1} + \underline{w_2}) &= \underline{v} \wedge \underline{w_1} + \underline{v} \wedge \underline{w_2} \\
(s\underline{v}) \wedge \underline{w} &= s(\underline{v} \wedge \underline{w}) \\
\underline{v} \wedge (s\underline{w}) &= s(\underline{v} \wedge \underline{w}) \\
\underline{v} \wedge \underline{v} &= \underline{0}
\end{aligned}$$

wherever they make sense.

EExercise 212. (on Sheet 18) Consider making a "tent" as shown. Note that unlike the "tent" shape used in Example 9.10, here the base of the tent is "upside down" compared to the roof (which is more like how you orient the fabric in a real tent!) and we have filled in the sides.



Suppose we have vectors $\underline{u}$, $\underline{v}$, and $\underline{w}$ as shown:



Now we can define elements of $\mathbb{R}^3 \wedge \mathbb{R}^3$ modelling all the sides as follows:

- The left side is $\underline{u} \wedge \underline{v}$
- The right side is $\underline{u} \wedge \underline{w}$
- The floor is $-(\underline{u} \wedge (\underline{v} + \underline{w}))$ (because it's "upside down")
- The back face is $(1/2)(\underline{v} \wedge \underline{w})$

- The front face is $-(1/2)(\underline{v} \wedge \underline{w})$

What do you expect the sum of all these vectors in $\mathbb{R}^3 \wedge \mathbb{R}^3$ is? Prove your answer correct. When justifying your steps, you can use "definition of $\mathbb{R}^3 \wedge \mathbb{R}^3$" to justify the use of any of the equations listed for this vector space.

9.2 Linear transformations

9.2.1 Linear transformations

Definition 81: linear transformation

Let K be a field and V and W be vector spaces over K. A function $f : V \to W$ is called a **linear transformation** (or **linear function**) if and only if for all $\underline{v_1}, \underline{v_2} \in V$ and r in K we have

- $f(\underline{v_1} + \underline{v_2}) = f(\underline{v_1}) + f(\underline{v_2})$ and
- $f(r\underline{v_1}) = rf(\underline{v_1})$

Note that this is somewhat more restrictive than you might have been expecting. In particular a linear transformation must send the $\underline{0}$ vector of the source vector space to the $\underline{0}$ vector of the output space.

CExercise 213. (on Sheet 19) Let K be a field and V and W be vector spaces over K, and let $f : V \to W$ be a linear transformation. Prove the following:

(i) $f(\underline{0}) = \underline{0}$

(ii) for all $\underline{v} \in V$, the multiplicative inverse of $f(\underline{v})$ is $f(-\underline{v})$.

This may be confusing if you are used to using the term "linear polynomial" to mean a function from $\mathbb{R}$ to $\mathbb{R}$ (or $\mathbb{C}$ to $\mathbb{C}$) which sends its input x to something of the form $ax + b$. However, the definition of linear function is more restrictive.

Example 9.24. (Non-example) Let us for the moment continue using the confusing terminology "linear polynomial" as above. Then a "linear polynomial" $x \mapsto ax + b$ is a linear function if and **only** if $b = 0$.

For the purposes of this unit, it is better to avoid this confusing terminology when discussing linear algebra. Note that this makes linear transformations a very simple concept when they go between 1-dimensional vector spaces. We shall explore this shortly.

Example 9.25. For any vector sapce V, the identity function is a linear transformation.

Example 9.26. Let U, V, and W be vector spaces over the same field, and $g : U \to V$ and $f : V \to W$ be linear transformations, then the composite function $fg : U \to W$ is a linear transformation. Let's check the two properties; let $\underline{v_1}, \underline{v_2} \in U$ and r be a scalar.

- We check preservation of vector addition

$$\begin{aligned}
(fg)(\underline{v_1} + \underline{v_2}) &= f(g(\underline{v_1} + \underline{v_2})) && \text{def of function composition} \\
&= f(g(\underline{v_1}) + g(\underline{v_2})) && \text{linearity of } g \\
&= f(g(\underline{v_1})) + f(g(\underline{v_2})) && \text{linearity of } f \\
&= (fg)(\underline{v_1}) + (fg)(\underline{v_2}) && \text{def. function composition}
\end{aligned}$$

- We check preservation of scalar multiplication

$$\begin{aligned}(fg)(r\underline{v_1}) &= f(g(r\underline{v_1})) && \text{def. function composition}\\ &= f(rg(\underline{v_1})) && \text{linearity of } g\\ &= rf(g(\underline{v_1})) && \text{linearity of } f\\ &= r(fg)(\underline{v_1}) && \text{def. function composition}\end{aligned}$$

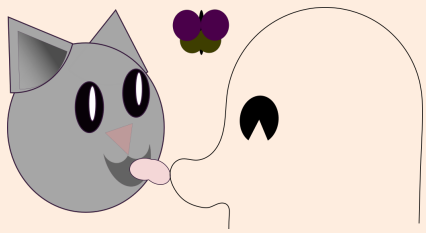
Example 9.27. For any vector space V and scalar r, the function $s_r : V \to V$ which sends $\underline{v}$ to $r\underline{v}$ is a linear transformation. We call this a "uniform scaling".

Example 9.28. **(Non-example!)** For any vector space V and vector $\underline{w}$, the function from $V \to V$ which sends a vector $\underline{v}$ to $\underline{v} + \underline{w}$ is usually **not** a linear transformation. The only exception is when $\underline{w} = \underline{0}$, in which case this is the identity transformation.

Example 9.29. Let V and W be vector spaces over the same field. Then there is a linear function $i_0 : V \to V \oplus W$ which sends $\underline{v} \in V$ to $(\underline{v}, \underline{0})$ in $W \oplus W$. Similarly there is a linear transformation $i_1 : W \to V \oplus W$ which sends $\underline{w}$ to $(\underline{0}, \underline{w})$.

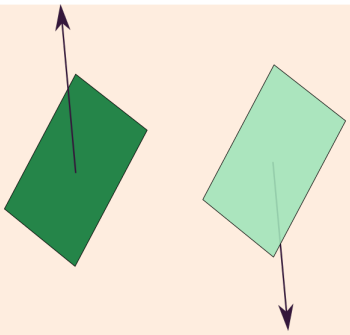
Example 9.30. Let V and W be vector spaces over the same field. Then there is a linear function $\pi_0 : V \oplus W \to V$ which sends $(\underline{v}, \underline{w})$ in $W \oplus W$ to $\underline{v}$. Similarly there is a linear transformation $\pi_1 : V \oplus W \to W$ which sends $(\underline{v}, \underline{w})$ to $\underline{w}$.

EExercise 214. (on Sheet 19) Suppose that while a cat is licking your nose, it is distracted by a butterfly and pauses licking, causing your nose to be held still, but with an upward deflection as shown in the figure:



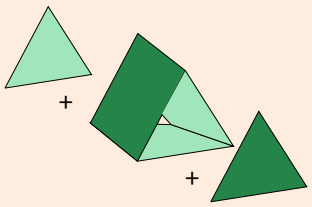
You can simulate this by moving the tip of your nose upwards with the palm of your hand using only friction — don't push inwards too much. You feel mechanical stress in your nose: if it were to be released it would spring back into its usual position, but it is held upwards by the force from the cat's tongue (or your palm) — on which it exerts an equal and opposite force.

If we consider a tiny polygonal surface inside your nose, then the distortion of your nose means a force is being applied to it. Note that this could be a friction force which applies more 'along' the surface; it doesn't just have to be a push going 'through' the surface. We consider the function f which maps a polygon in $\mathbb{R}^3 \wedge \mathbb{R}^3$ to a force vector in $\mathbb{R}^3$: the force applied to the surface by the material on the side the surface is facing.



The material on the other side must be applying an equal and opposite force, so for a polygon $\underline{p} \in \mathbb{R}^3 \wedge \mathbb{R}^3$ we must have $f(-\underline{p}) = -f(\underline{p})$ as depicted above. And a greater area polygon feels a proportionally greater force, so we have $f(r\underline{p}) = rf(\underline{p})$ for all $r \in \mathbb{R}$.

Now if we consider any small enclosed "tent-shaped" volume inside your nose:



Since your nose is not moving, the sum of the forces on the sides of the enclosed volume must be zero (i.e. it must be $\underline{0} \in \mathbb{R}^3$).

Prove that f is linear.

You may find it useful to choose names for some of the vectors in the figure, similar to those in eExercise 212. Note that you need only check the behaviour of f with respect to vector addition, as the behaviour with respect to scalar multiplication is given above. You may use the fact that every pair of vectors in $\mathbb{R}^3 \wedge \mathbb{R}^3$ can be assumed to be in the form of two "rectangles" with a common side forming the top of such a "tent" without proving it. Algebraically this means that you can assume that for all $\underline{p}.\underline{q} \in \mathbb{R}^3 \wedge \mathbb{R}^3$ there exist vectors $\underline{u}, \underline{v}, \underline{w} \in \mathbb{R}^3$ such that $\underline{p} = \underline{u} \wedge \underline{v}$ and $\underline{q} = \underline{u} \wedge \underline{w}$.

The ideas in the exercise above don't work if we consider a moving object – e.g. if the cat was actively accelerating your nose! However this is a situation where linear algebra applies approximately to very small things. If we consider tiny polygons inside an accelerating nose, then when we put them together in a "tent" shape, the volume of the tent is tiny. That means the mass is tiny, and hence the force required for the cat to accelerate that tiny bit of nose is also tiny. Engineers can treat this tiny force as zero. Then your answer to the exercise above shows that the force function is approximately linear! You might wonder whether this is trivial because if the faces are tiny, then you might think the force applied to each face must be tiny too. But consider a rocket accelerating a single piece of spaghetti:

It doesn't take much force to accelerate a tiny *volume* within the spaghetto, but if it is extremely long, the force through a tiny *surface* may have to be big enough to accelerate all the mass on one side, so could be very large. So engineers can pretend one is zero but *not* the other!

The space of linear transformations

The set of all linear transformations between V and W form a vector space, indeed, a subspace of the space of all functions.

Proposition 9.9

Let K be a field and V and W be vector spaces over K. Recall from Exercise 200 that the set of all functions from V to W can be given the structure of a vector space, in which:

- $(\underline{f} + \underline{g})(x) = \underline{f}(x) + \underline{g}(x)$
- $(r\underline{f})(x) = r(\underline{f}(x))$

The set of linear transformations from V to W form a subspace of this vector space.

Proof. We must show that the set of linear transformations is closed under the addition and scalar multiplication given. For all linear transformations $\underline{f}, \underline{g} : V \to W$ thought of as vectors in the vector space of functions from V to W, $\underline{v_1}, \underline{v_2}, \underline{v} \in V$ and scalars r, s we have:

- To show that linear transformations are closed under the $+$ operation of the space of functions into W we calculate

$$\begin{aligned}
(\underline{f} + \underline{g})(\underline{v_1} + \underline{v_2}) &= \underline{f}(\underline{v_1} + \underline{v_2}) + \underline{g}(\underline{v_1} + \underline{v_2}) && \text{def } + \text{ in space of functions} \\
&= \underline{f}(\underline{v_1}) + \underline{f}(\underline{v_2}) + \underline{g}(\underline{v_1}) + \underline{g}(\underline{v_2}) && \text{linearity of } f \text{ and } g \\
&= (\underline{f}(\underline{v_1}) + \underline{g}(\underline{v_1})) + (\underline{f}(\underline{v_2}) + \underline{g}(\underline{v_2})) && \text{assoc. comm. of } + \text{ in } W \\
&= (\underline{f} + \underline{g})(\underline{v_1}) + (\underline{f} + \underline{g})(\underline{v_2}) && \text{def } +
\end{aligned}$$

  and

$$\begin{aligned}
(\underline{f} + \underline{g})(r\underline{v_1}) &= \underline{f}(r\underline{v}) + \underline{g}(r\underline{v}) && \text{def. } + \text{ in function space} \\
&= r(\underline{f}(\underline{v})) + r(\underline{g}(\underline{v})) && \text{linearity of } f \text{ and } g \\
&= r(\underline{f}(\underline{v}) + \underline{g}(\underline{v})) && \text{vector distr. in } W \\
&= r((\underline{f} + \underline{g})(\underline{w})) && \text{def. } + \text{ in function space}
\end{aligned}$$

  So $\underline{f} + \underline{g}$ is a linear transformation, and hence linear transformations are closed under $+$ in the function space.

- To show that linear transformations are closed under scalar multiplication in the function space we calculate

$$\begin{aligned}
(r\underline{f})(\underline{v_1} + \underline{v_2}) &= r(\underline{f}(\underline{v_1} + \underline{v2})) && \text{def. scalar mult. in function space} \\
&= r(\underline{f}(\underline{v_1}) + \underline{f}(\underline{v_2})) && \text{linearity of } f \\
&= r\underline{f}(\underline{v_1}) + r\underline{f}(\underline{v_2}) && \text{vector distr. in } W \\
&= (rf)(\underline{v_1}) + (rf)(\underline{v_2}) && \text{def. scalar mult. in function space}
\end{aligned}$$

  and

$$(r\underline{f})(s\underline{v}) = r\underline{f}(s\underline{v}) \qquad \text{def. scalar mult. in function space}$$

$$
\begin{aligned}
&= sr\underline{f}\underline{v} && \text{linearity of } f\text{, calc. with scalars} \\
&= s((r\underline{f})(\underline{v})) && \text{def. scalar mult. in function space.}
\end{aligned}
$$

So $r\underline{f}$ is a linear transformation, and hence linear transformations are closed under scalar multiplication in the function space.

That means linear transformations form a subspace just as we said.

This vector space is important, so we define its traditional name:

Definition 82: vector space of linear transformations, 'Hom'

Let K be a field and V and W be vector spaces over K. Then the space of linear transformations from V to W is called $Hom(V, W)$.

The name "Hom" comes from the Greek "homomorphism" ("similar shape") which you can think of as being because linear transformations send "straight lines" to 'straight lines" and thus preserve something about the "shape" of things in their inputs. (Though they may also distort shapes in various ways!)

The most important example of a space of linear transformations is when the target space is the field itself (with its standard addition and multiplication). We call this the **dual space** of the source space.

Definition 83: dual space

Let V be a vector space over the field K. Then the **dual space** $V^\star$ is the space $Hom(V, K)$

This gives rise to an important class of examples of linear transformations. Given any linear transformation between vector spaces, we get a linear transformation going *in the other direction* between their dual spaces, given by function composition.

Example 9.31. Let V and W be vector spaces over the same field. Let $f : V \to W$ be a linear transformation. Then the function $f^\star : W^\star \to V^\star$ given by $(f^\star)\underline{g} = \underline{g} \circ f$.

Let's see that this gives us something in $V^\star$, i.e. a linear transformation from V to K. We know that $f : V \to W$ is a linear transformation and $\underline{g} \in W^\star = Hom(W, K)$ so $\underline{g} : W \to K$ is also a linear transformation. Then the composite is a linear transformation by Example 9.26. We can picture the situation as follows:

$$
\begin{tikzcd}
V \arrow[rr, "f"] \arrow[dr, "\underline{g}\circ f"'] & & W \arrow[dl, "\underline{g}"] \\
& K &
\end{tikzcd}
$$

Now we should check that this is really a linear transformation. Let $\underline{g}, \underline{h} \in W^\star$, $\underline{v} \in V$, and r a scalar. We look at the two conditions.

- We check preservation of vector addition:

$$
\begin{aligned}
f^\star(\underline{g} + \underline{h})(\underline{v}) &= ((\underline{g} + \underline{h}) \circ f)\underline{v} && \text{def. } f^\star \\
&= (\underline{g} + \underline{h})(f(\underline{v})) && \text{def. function comp.} \\
&= \underline{g}(f(\underline{v})) + \underline{h}(f(\underline{v})) && \text{def. } + \text{ in the function space}
\end{aligned}
$$

$$
\begin{aligned}
&= (\underline{g} \circ f)(\underline{v}) + (\underline{h} \circ f))(\underline{v}) && \text{def. function comp.} \\
&= (f^\star(g))(\underline{v}) + (f^\star(h)) && \text{def. } f^\star
\end{aligned}
$$

- We check preservation of scalar multiplication:

$$
\begin{aligned}
f^\star(r\underline{g})(\underline{v}) &= ((r\underline{g}) \circ f)(\underline{v}) && \text{def. } f^\star \\
&= (r\underline{g})(f(\underline{v})) && \text{def. function comp.} \\
&= r(\underline{g}(f(\underline{v})) && \text{def. scalar mult. in function space} \\
&= r((\underline{g} \circ f)(\underline{v})) && \text{def. function comp.} \\
&= r((f^\star(f))(\underline{v})) && \text{def. } f^\star
\end{aligned}
$$

So putting these together we see that this is indeed an example of a linear transformation!

Subspaces associated to a linear transformation

Given a linear transformation $f : V \to W$ we are interested in its **image**, a subspace of the target W, and its **kernel**, a subspace of the source V. Here we define these subsets and prove that they are subspaces.

Definition 84: image of a linear transformation

Let V and W be vector spaces over the same field, and f be a linear transformation from V to W. Then we let

$$Im(f) = \{\underline{w} \in W \mid \text{there exists } \underline{v} \in V \text{ such that } f(\underline{v}) = \underline{w}\}$$

We call this the **image** of f.

Proposition 9.10

Let V and W be vector spaces over the same field, and f be a linear transformation from V to W. Then $Im(f)$ is a subspace of W.

Proof. Suppose $\underline{w_1}, \underline{w_2} \in Im(f)$. Then there exist $\underline{v_1}, \underline{v_2} \in V$ such that $f(\underline{v_1}) = \underline{w_1}$ and $f(\underline{v_2}) = \underline{w_2}$. Then by linearity of f we have $f(\underline{v_1} + \underline{v_2}) = \underline{w_1} + \underline{w_2}$ so we have found an element of V mapped to the sum as required.

Similarly, suppose $\underline{W} \in Im(f)$ then there is $\underline{v} \in V$ with $f(\underline{v}) = \underline{w}$. Now for a scalar r we have $f(r\underline{v}) = r\underline{w}$ by linearity of f, so the scalar multiplication of r with $\underline{w}$ is in Imf as required.

Thinking of which subspace of V we are interested in is more subtle. The image of f tells us how much of the structure of W f can "see". We are also interested in how much detail about V is lost by applying f, and the way to study this is to look at the set of vectors which f sends to $\underline{0}$.

Definition 85

Let V and W be vector spaces over the same field, and f be a linear transformation from V to W. Then we let

$$Ker(f) = \{\underline{v} \in V \mid f(\underline{v}) = \underline{0}\}$$

We call this the **kernel** of f.

Proposition 9.11

Let V and W be vector spaces over the same field, and f be a linear transformation from V to W. Then $Ker(f)$ is a subspace of W.

Proof. Exercise.

CExercise 215. (on Sheet 19) Prove Proposition 9.11.

One reason why these spaces are important is that their dimensions tell us about the linear transformation in question.

Proposition 9.12

Let V and W be finite dimensional vector spaces over the same field, and f be a linear transformation from V to W. Then f is surjective if and only if the dimension of Imf is equal to the dimension of the target space W.

We don't go into the proof of this Proposition since it requires results about bases beyond those we proved in the previous section!

On the other hand, it is much more straightforward to show the following.

Proposition 9.13

Let V and W be finite dimensional vector spaces over the same field, and f be a linear transformation from V to W. Then f injective if and only if $Kerf = \{\underline{0}\}$

Proof. Exercise!

EExercise 216. (on Sheet 19) Prove Proposition 9.13. Remember to do both directions of the "if and only if"!

9.2.2 Matrices

In this section we study how linear transformations interact with a basis.

Definition 86

Let V and W be finite dimensional vector spaces over the same field and $f : V \to W$ be a linear transformation. Let $B = \underline{b_1}, ..., \underline{b_m}$ be a basis for V and $C = \underline{c_1}, ..., \underline{c_n}$ be a basis for W then the **matrix** M of f with respect to B and C is an indexed set of scalars $\{M_{i,j} \mid i \in \{1, ..., n\}, j \in \{1, ..., m\}\}$ such that $M_{i,j}$ is the $\underline{c_i}$ component of the vector $f(\underline{b_j})$. It is usually displayed as a rectangular grid of numbers with n rows and m columns. We say that this is an n by m matrix.

Example 9.32. Let's look at the example of a uniform scaling. Consider the vector space $\mathbb{R}^3$. We choose any basis $B = \underline{u_1}, \underline{u_2}, \underline{u_3}$, and now we can express vectors using the column notation with respect to this basis. In particular we have

$$\underline{u_1} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}_B, \underline{u_2} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_B, \underline{u_3} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_B.$$

When we apply the uniform scaling to these, we obtain:

$$s_r(\underline{u_1}) = \begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix}_B, s_r(\underline{u_2}) = \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix}_B, s_r(\underline{u_3}) = \begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix}_B.$$

So to arrange these in a rectangular grid to get the usual notation for the matrix, we first put them side-by-side:

$$\begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix}_B \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix}_B \begin{pmatrix} 0 \\ 0 \\ t \end{pmatrix}_B.$$

Then remove the brackets, and just bracket the whole thing:

$$\begin{pmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{pmatrix}$$

Note that in the matrix notation we will typically leave the bases of the vector spaces involved implicit, but if we want to we can specify them like so:

$$\begin{pmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{pmatrix}_{B,B}$$

Note that in this example to keep things simple, we have picked the same basis for V when it is the source space, and when it is the target space.

Example 9.33. Let's see what the example above looks like if we use a slightly different basis for V when it is the target space, but keep the same basis when it is the source space. This is somewhat unnatural, as one usually chooses the same basis when the source and target are the same. However as we saw last week, it is possible to describe "the same" vector space in lots of different ways when it comes from different application areas, and we could have a situation where the "natural" basis for the intended application for the source space is different to that of the target space.

Returning to the example above, let $B' = \underline{u_1}, \underline{u_2}, (\underline{u_1} + \underline{u_3})$. You should check this is a basis; most of the work has been done already by Proposition 9.4.

Now

$$\begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix}_B = \begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix}_{B'} \text{ and } \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix}_B = \begin{pmatrix} 0 \\ r \\ 0 \end{pmatrix}_{B'}$$

as the corresponding basis vectors are the same (you could check this by using the

uniqueness of the basis representation of a vector). But the vector

$$\begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix}_B = r\underline{u_3}$$

expressed in the new basis is $r(\underline{u_3} + \underline{u_1}) - r\underline{u_1}$ so we have

$$\begin{pmatrix} 0 \\ 0 \\ r \end{pmatrix}_B = \begin{pmatrix} -r \\ 0 \\ r \end{pmatrix}_{B'}$$

Which means we have

$$\begin{pmatrix} r & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & r \end{pmatrix}_{B,B} = \begin{pmatrix} r & 0 & -r \\ 0 & r & 0 \\ 0 & 0 & r \end{pmatrix}_{B,B'}$$

This emphasises that the matrix corresponding to a linear transformation is a basis-dependent notion.

Exercise 217. Which matrix corresponds to the identity function on $\mathbb{R}^3$?

Example 9.34. Consider $\mathbb{R}^2 = \mathbb{R} \oplus \mathbb{R}$. The function $i_0 : \mathbb{R} \to \mathbb{R}^2$ defined in Example 9.29 corresponds to the matrix

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Whereas i_1 corresponds to

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

.

Meanwhile π_0 defined in Example 9.30 is given by

$$\begin{pmatrix} 1 & 0 \end{pmatrix}$$

while π_1 s given by

$$\begin{pmatrix} 0 & 1 \end{pmatrix}.$$

Two properties make the matrix notation important. The first is that the linear transformations between two spaces are in one-to-one correspondence with the matrices of the appropriate type.

Lemma 9.14

Let V and W be vector spaces over a field K, and let B be a basis for V. Let $f, g : V \to W$ be linear transformations. Suppose that for all $\underline{b} \in B$ we have $f(\underline{b}) = g(\underline{b})$. Then $f = g$.

Proof. Let $\underline{v} \in V$. We need so show that $f(\underline{v}) = g(\underline{v})$. Since B is a basis, there exists a unique subset $\{\underline{b_1}, ...\underline{b_n}\}$ of B and scalars $\{r_1, ...r_n\} \subseteq K$ such that $\underline{v} = r_1\underline{b_1} + ... + r_n\underline{b_n}$.

Then since f is linear we find that

$$\begin{aligned} f(\underline{v}) &= r_1 f(\underline{b_1}) + r_n f(\underline{b_n}) && \text{linearity of } f \\ &= r_1 g(\underline{b_1}) + r_n g(\underline{b_n}) && f \text{ and } g \text{ agree on basis vectors} \\ &= g(\underline{v}) && \text{linearity of } g \end{aligned}$$

Theorem 9.15

Let V and W be finite dimensional vector spaces over a field K. Let $B = \underline{b_1}, ..\underline{b_m}$ be a basis for V and $C = \underline{c_1}, ..., \underline{c_n}$ a basis for W. Then every indexed set $M_{i,j} \mid i \in \{1, ..., m\}, j \in \{1, ..., n\}$ is the matrix of a unique linear transformation with respect to the bases B and C

Proof. Given a vector

$$\underline{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}_B \in V$$

we define

$$f(\underline{x}) = x_1 \begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + ... + x_m \begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C$$

We need to show that this is linear, that the matrix of the linear transformation is M and that it is the only linear transformation with this matrix.

First, linearity.

$$\begin{aligned} & f\left(\begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}_B + \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}_B\right) \\ =& f\begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_m + y_m \end{pmatrix}_B && \text{by Theorem 9.7} \\ =& (x_1 + y_1)\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + ... + (x_m + y_m)\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C && \text{def. } f \\ =& x_1\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + y_1\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + ... + x_m\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C + y_m\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C && \text{vector distr.} \\ =& x_1\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + ... + x_m\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C + y_1\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_C + ... + y_m\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_C && + \text{comm.} \end{aligned}$$

$$=f\begin{pmatrix}x_1\\ \vdots\\ x_m\end{pmatrix}_B+f\begin{pmatrix}y_1\\ \vdots\\ y_m\end{pmatrix}_B \qquad \text{def. } f$$

and similarly

$$\begin{aligned}
&f\left(r\begin{pmatrix}x_1\\ \vdots\\ x+m\end{pmatrix}_B\right)\\
=&f\begin{pmatrix}rx_1\\ \vdots\\ rx_m\end{pmatrix}_B && \text{by Theorem 9.7}\\
=&rx_1\begin{pmatrix}M_{1,1}\\ \vdots\\ M_{1,n}\end{pmatrix}_C+\ldots+rx_m\begin{pmatrix}M_{m,1}\\ \vdots\\ M_{m,n}\end{pmatrix}_C && \text{def. } f\\
=&r\left(x_1\begin{pmatrix}M_{1,1}\\ \vdots\\ M_{1,n}\end{pmatrix}_C+\ldots+x_m\begin{pmatrix}M_{m,1}\\ \vdots\\ M_{m,n}\end{pmatrix}_C\right) && \text{vector distr.}\\
=&rf\begin{pmatrix}rx_1\\ \vdots\\ rx_m\end{pmatrix}_B && \text{def. } f
\end{aligned}$$

Which takes care of linearity.

To see that the matrix of f is M is straightforward from the definition, using the fact that a basis vector $\underline{b_i}$ has coefficients which are 0 for all other basis vectors and 1 for itself. So f maps it to

$$\begin{pmatrix}M_{i,1}\\ \vdots\\ M_{i,n}\end{pmatrix}_C$$

as required.

Finally since we have shown this function is linear, we may appeal to Lemma 9.14 for its uniqueness.

The notation becomes rather cumbersome in the proof above, although the column notation does help with readability to some extent. Note that wherever it is possible to stay aloof from co-ordinate calculation and apply one of the properties of a vector space, it is usually best to do so: we could have kept calculating in the middle of these proofs until we had just a single column vector, then applied properties of scalars — but this would have resulted in an even more illegible proof!

The second property which makes the matrix notation important is that it interacts well with function composition.

Theorem 9.16

Let U, V, W be finite dimensional vector spaces over K and $B = \underline{b_1}, \ldots, \underline{b_l}$, $C = \underline{c_1}, \ldots, \underline{c_m}$ and $D = \underline{d_1}, \ldots \underline{d_n}$ be bases for U, V, W respectively. Let $f : U \to V$ with matrix L and

$g : V \to W$ with matrix M. Then $g \circ f$ has matrix N where

$$N_{i,k} = \sum_{j=1}^{m} M_{i,j} L_{j,k}$$

Proof. Let consider $\underline{b_k}$. By the definition of L,

$$f(\underline{b_k}) = L_{i,1}\underline{c_1} + \ldots + L_{i,m}\underline{c_m}$$

so by the definition of M

$$g(f(\underline{b_i})) = L_{i,1}\begin{pmatrix} M_{1,1} \\ \vdots \\ M_{1,n} \end{pmatrix}_D + \ldots + L_{i,m}\begin{pmatrix} M_{m,1} \\ \vdots \\ M_{m,n} \end{pmatrix}_D$$

The k'th component of which is

$$\sum_{j=1}^{m} M_{i,j} L_{j,k}$$

by Theorem 9.7, as required.

This operation on matrices is called the **product of matrices**, and we write ML for the product of matrices M and L as in the theorem. Note that for the product to be well defined, the number of columns of M must be equal to the number of rows of L. After all, these must both be the dimension of the same space (B in the theorem).

Exercise 218. Prove that the product of matrices is associative.

Definition 87

The **transpose** of a matrix $M = \{M_{i,j} \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$ is the matrix $M^T = \{M_{j,i} \mid i \in \{1, \ldots, m\}, j \in \{1, \ldots, n\}\}$.

Note that the indices have flipped over, so this matrix is obtained by flipping the matrix M over the diagonal running from the top left to the bottom right. If M represents the linear transformation f with respect to come basis, then M^T represents $f^\star$ with respect to the same basis.

CExercise 219. (on Sheet 19)

Let

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$N = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$P = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$$

Consider each of these possible matrix products. In each case, if the product is not defined, say why, and if it is defined, calculate it.

(i) MN

(ii) NM

(iii) M^TN

(iv) MN^T

(v) MP

(vi) M^TP

(vii) MP^T

(viii) NMP^T

(ix) MP^TN

Geometric examples of linear transformations

For linear transformations from $\mathbb{R}^2$ to $\mathbb{R}^2$ or from $\mathbb{R}^3$ to $\mathbb{R}^3$, there are many linear transformations which we can visualise in a nice geometric way. The idea is that we pick a basis for these vector spaces, and visualise the basis vectors as arrows of the same length pointing in orthogonal directions, like the axes of a graph. Then the following matrices have the following geometric meanings.

Example 9.35. As well as uniform scalings, we can scale by one scale factor in one direction, and another in the other direction. This gives rise to a non-uniform scaling matrix of the form

$$\begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix}$$

Example 9.36. A shear transformation parallel to one axis is represented by the matrix

$$\begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}$$

while a shear parallel to the other axis is represented by

$$\begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix}$$

Exercise 220. Since you may be less familiar with the idea of a shear transformation, pick several vectors in $\mathbb{R}^2$ and apply the transformations above when $r = 2$ to see what happens.

Example 9.37. A reflection in one axis is given the the matrix

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

and in the other by

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Example 9.38. A rotation anticlockwise by an angle θ is represented by the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Some concrete examples are given by angles which have "nice" sines and cosines. For example a rotation by $\pi/6$ is given by the matrix

$$\begin{pmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}$$

Whereas a rotation by $\pi/4$ is given by the matrix

$$\begin{pmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{pmatrix}$$

CExercise 221. (on Sheet 19) The rotation matrix for an angle of $\pi/2$ (that is, a $90°$ rotation) is even simpler than those above. However in this exercise we complicate things by considering what happens in 3D. Here we must decide which axis we are rotating around. Consider a basis $\underline{b_1}, \underline{b_2}, \underline{b_3}$ we wish to rotate by $\pi/2$ around the line which each basis vector lies in.

- Give matrices for a $\pi/2$ rotation around each of the basis vectors $\underline{b_1}, \underline{b_2}, \underline{b_3}$.
- Using these matrices, show that matrix multiplication is not commutative.

COMP11120, Semester 2
Assessed Exercise Sheet 18

Core exercises

- Complete cExercise 196
- Complete cExercise 201
- Complete cExercise 209

Extensional exercises

- Complete eExercise 205
- Complete eExercise 212

COMP11120, Semester 2
Assessed Exercise Sheet 19

Core exercises

- Complete cExercise 213
- Complete cExercise 215
- Complete cExercise 219 and cExercise 221

Extensional exercises

- Complete eExercise 214
- Complete eExercise 216